Population Group-Wise Bank Deposits Analysis Using Python

1. Project Overview

This project focuses on analyzing the dataset using Python to extract meaningful financial and demographic insights regarding banking penetration and deposit trends. The analysis includes data cleaning, exploratory data analysis (EDA), statistical evaluation, and visual representations to understand patterns across different states, regions, and population groups (Rural, Semi-Urban, Urban, Metropolitan).

The project aims to transform raw banking data into actionable information that supports better regional financial planning and resource allocation.

2. Data Source

- **Dataset Name:** Population Group-Wise Deposits Dataset
- **Source:** CSV File / Reserve Bank of India (RBI) / Open Government Data
- **File Type:** CSV
- **Number of Records:** 14,968
- **Number of Attributes:** 11

The dataset contains regional and demographic banking metrics, including the number of bank offices, number of accounts, and total deposit amounts across different years, states, districts, and population categories.

3. Problem Statement

Financial institutions and policymakers collect massive quantities of transactional and regional data, but often struggle to understand how capital is distributed across different population groups and geographical regions. Without comprehensive data analysis, identifying underserved regions or evaluating deposit concentration trends remains difficult.

This project aims to analyze the population group-wise deposit dataset to identify key drivers of deposit volumes, banking office distributions, and account penetration across India's diverse geographic and demographic segments.

4. Objectives

- To understand the dataset structure and data types.
- To clean and prepare the data for regional and statistical analysis.
- To handle missing values and validate duplicate records.
- To perform Exploratory Data Analysis (EDA) across years and regions.
- To identify patterns and trends in deposit amounts and office distributions.
- To create meaningful visualizations illustrating disparities across population groups.
- To generate actionable insights for financial inclusion and banking expansion.

5. Attribute (Column / Feature) Details

Attribute Name	Description
id	Unique identifier for each record
year	Year of the recorded data
state_name	Name of the state or union territory
state_code	Numerical code assigned to the state
district_name	Name of the district
district_code	Numerical code assigned to the district
region	Geographic region (e.g., Eastern, Southern, Northern, etc.)
population_group	Classification of population density (Rural, Semi-Urban, Urban, Metropolitan)
no_of_offices	Total number of bank offices/branches
no_of_accounts	Total number of deposit accounts
deposit_amount	Total amount of deposits

6. Tools & Technologies

- **Python**
- **Pandas** (for data manipulation and aggregation)
- **NumPy** (for numerical operations)
- **Matplotlib** (for plotting charts)
- **Seaborn** (for statistical data visualization)
- **Jupyter Notebook / Google Colab**

7. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using standard data preprocessing techniques:

Data Cleaning Steps

- **Checked data types** of all columns (`id`, `year`, `state_code`, `district_code` as integers; metrics as floats; text as objects).
- **Identified missing values** in `no_of_offices`, `no_of_accounts`, and `deposit_amount` (607 null records identified).
- **Handled null values** using appropriate imputation or filtering techniques depending on the analytical scope.
- **Removed duplicate records** to ensure data integrity.
- **Corrected inconsistent data** and verified state and district naming conventions.

Feature Engineering

New features and calculated metrics were derived to enhance analysis:

- **Average Deposit per Account:** Calculated as `deposit_amount / no_of_accounts` to measure account wealth density.
- **Accounts per Office:** Calculated as `no_of_accounts / no_of_offices` to gauge branch workload and customer footfall.
- **Deposit per Office:** Calculated as `deposit_amount / no_of_offices` to evaluate branch productivity.

8. Analysis and Visualizations

The dataset was examined using Exploratory Data Analysis (EDA) techniques to uncover underlying geographical, temporal, and demographic relationships.

9. Exploratory Data Analysis (EDA)

Initial inspection included:

- **DataFrame structure verification:** Validating column dimensions and non-null counts.
- **Statistical summary of numerical columns:** Inspecting metrics like mean, median, standard deviation, and maximum limits for deposits and accounts.

- **Null value verification:** Quantifying missing records across specific population groups.
- **Duplicate record validation:** Ensuring uniqueness of entry identifiers.

EDA established a clear baseline of distribution characteristics, showing wide variations between rural and metropolitan banking metrics.

10. Statistical Analysis

Statistical measures were used to summarize the dataset's central tendency and dispersion:

- **Mean:** Average deposit amounts and account sizes across sectors.
- **Median:** Midpoint values to avoid skewness caused by metropolitan outliers.
- **Mode:** Most frequently occurring categories for population groups and regions.
- **Minimum value & Maximum value:** Bounding the range of bank offices (3 to $2,915$) and deposit amounts.
- **Standard deviation:** Measuring the dispersion and variance in regional deposit volumes.

11. Data Visualization

1. Univariate Analysis

- **Distribution of Deposit Amounts (with image):** Shows right-skewed distribution where most districts hold lower aggregate deposits while a few major financial hubs hold massive volumes.
- **Distribution of Population Groups (with image):** Compares the frequency of records across Rural, Semi-Urban, Urban, and Metropolitan classifications.
- **Distribution of Bank Offices (with image):** Illustrates the concentration of bank branches across regions.

2. Bivariate Analysis

- **Population Group vs. Total Deposit Amount (with image):** Demonstrates that Metropolitan and Urban groups account for the overwhelming majority of total deposits compared to Rural areas.
- **Number of Offices vs. Deposit Amount (with image):** Highlights a strong positive correlation between branch presence and total capital collected.
- **Year vs. Total Deposit Amount (with image):** Tracks the longitudinal growth of aggregate bank deposits over the study period.

3. Multivariate Analysis

- **Region, Population Group, and Deposit Amount (with image):** Analyzes how deposit volumes differ across regions when controlled for population group classification.
- **Correlation Heatmap (with image):** Displays correlation coefficients among `no_of_offices`, `no_of_accounts`, and `deposit_amount`, revealing tight interdependence between branch network scale and deposit collection.

- **Combined Feature Analysis (with image):** Explores account density alongside deposit sizes across different state tiers.

12. Summary of Findings

- Metropolitan and urban sectors dominate total deposit volumes despite rural areas covering a larger share of geographical entries.
- A strong linear relationship exists between the number of bank offices and total deposit amounts.
- Average deposit sizes per account vary significantly between rural and metropolitan tiers, reflecting income and wealth disparities.
- Deposit mobilization has steadily expanded across all population groups over the recorded years.
- Certain regions display higher account-to-office ratios, indicating heavier branch utilization in specific urban centers.

13. Types of Analysis

1. Descriptive Analysis

Summarizes historical banking figures:

- Average number of accounts per district
- Total deposit distribution nationwide
- Regional counts of banking offices

2. Diagnostic Analysis

Explains underlying trends and variations:

- Why metropolitan centers accumulate disproportionately higher deposit amounts
- Factors influencing lower branch densities in specific rural districts

3. Predictive Analysis

Forecasts future financial metrics:

- Predicting future deposit growth based on historical temporal trends
- Forecasting branch expansion requirements for developing semi-urban zones

4. Prescriptive Analysis

Suggests data-driven policy and business actions:

- Recommending targeted financial inclusion initiatives for underserved rural sectors
- Optimizing branch placement strategies based on account density and deposit yield

14. Future Enhancements and Suggestions

- Incorporate credit and loan data alongside deposits to compute complete credit-deposit (CD) ratios.
- Apply advanced machine learning regression models to predict regional deposit potentials.
- Build interactive PowerBI or Streamlit dashboards for real-time monitoring by policymakers.
- Automate reporting pipelines to ingest newer RBI annual data releases seamlessly.

15. Conclusion

This project successfully analyzed the dataset using Python, translating raw banking statistics into clear structural insights. Rigorous data preprocessing, statistical evaluation, and visual storytelling provided a comprehensive view of how deposits and banking infrastructure are distributed across India's population groups. The findings serve as a valuable reference for optimizing resource allocation and enhancing financial inclusion strategies.